



City-Wide AI Traffic Intelligence Engine

Smart India Hackathon 2026 — Comprehensive Study Guide

1. Comprehensive Problem Statement

Modern cities are expanding at an unprecedented rate, bringing immense pressure onto urban road networks. While governments have installed thousands of CCTV cameras across traffic intersections, the monitoring process remains highly inefficient:

- **Human Limitation in Surveillance:** It is physically impossible for human operators to actively monitor thousands of camera feeds simultaneously to detect traffic anomalies, violations, or congestion.
- **Disconnected Camera Networks:** Cameras exist as isolated nodes. If a suspicious or speeding vehicle passes camera A, tracking it as it moves to camera B, C, and D is a highly manual, time-consuming forensic task.
- **Reactive vs. Proactive Response:** Traffic police currently rely on reactive measures (responding to accidents or gridlocks after they happen) because they lack real-time algorithmic alerts for sudden congestion spikes or unusual vehicle behaviors.

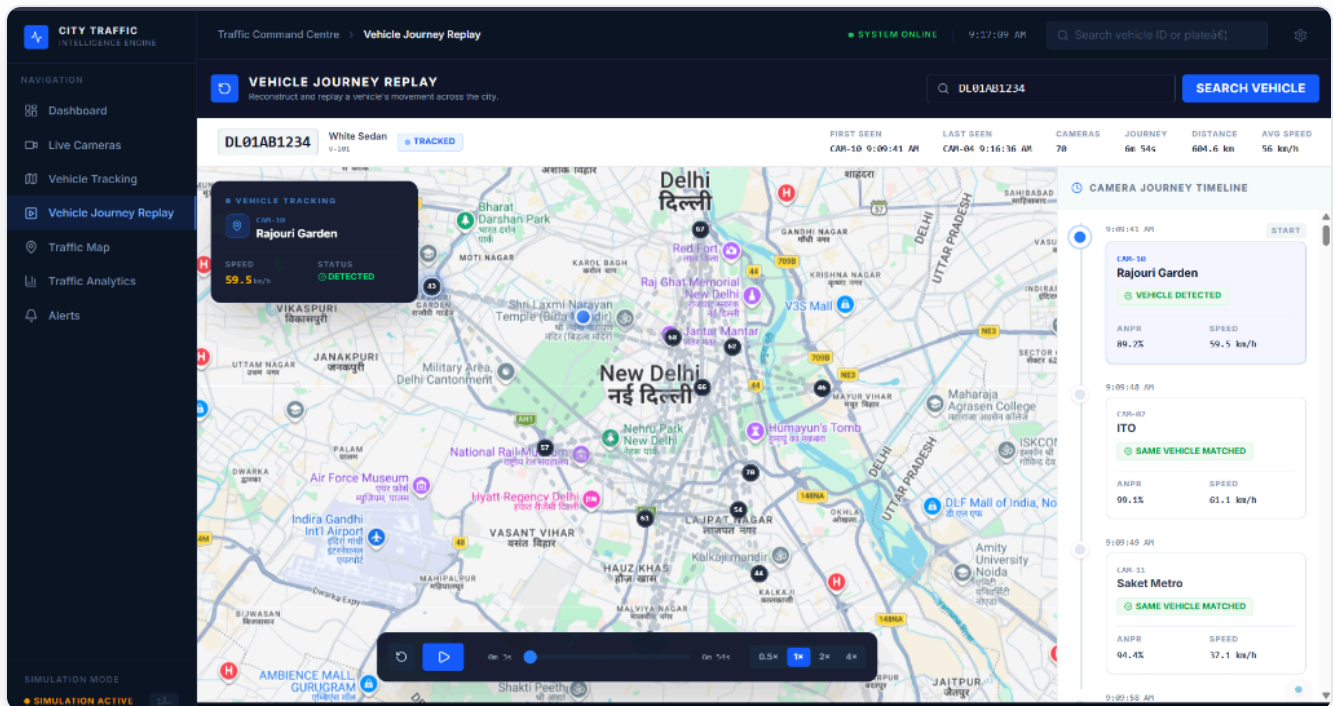
2. Our Solution & Core Features

We developed an enterprise-grade AI command center that unifies isolated CCTV cameras into a singular, intelligent tracking network. By processing video feeds at the edge, our software extracts actionable intelligence in real-time.

A. Cross-Camera Tracking & Journey Replay

How it works: When an operator inputs a vehicle's license plate, our backend correlates historical ANPR detections across the entire city grid. It deduplicates back-to-back detections at the same intersection and plots a clean trajectory.

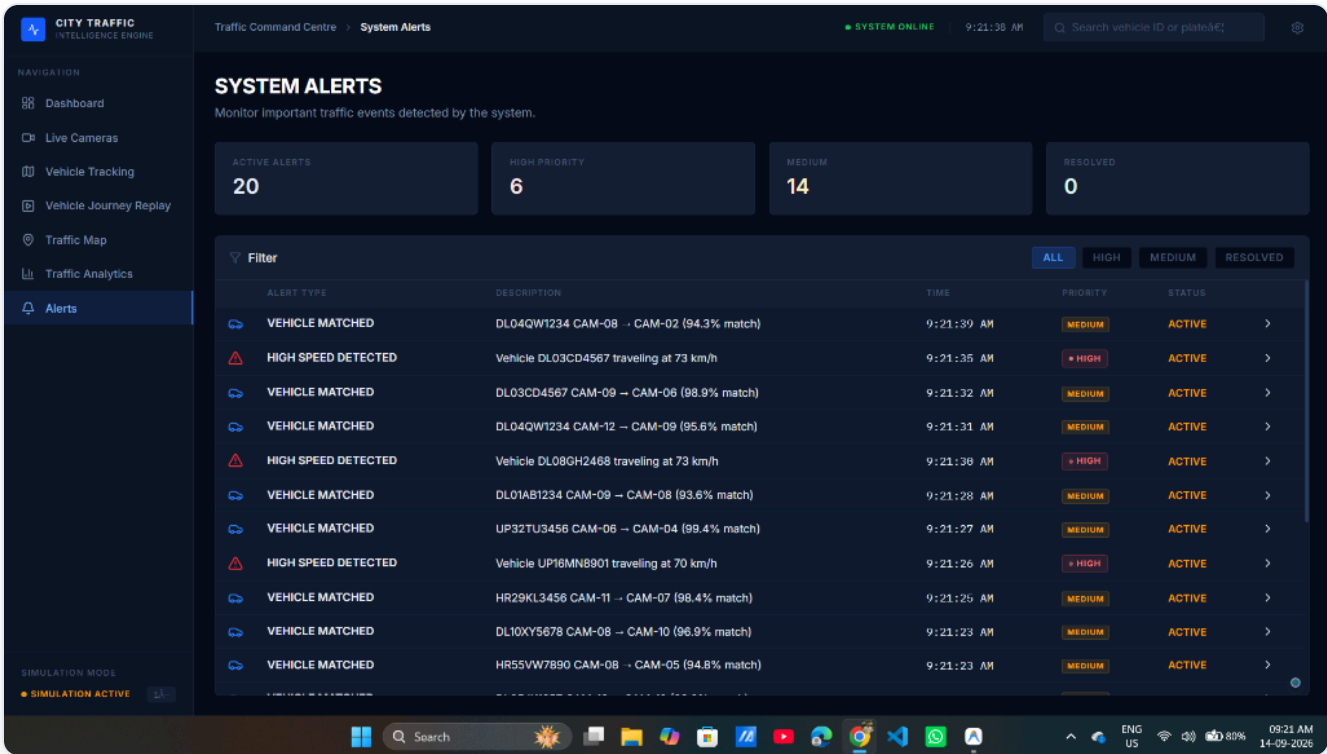
The Interface: A cinematic Google Maps integration with glowing blue trajectory trails and an auto-scrolling timeline log that mimics a live forensic replay.



B. Automated Alert Engine

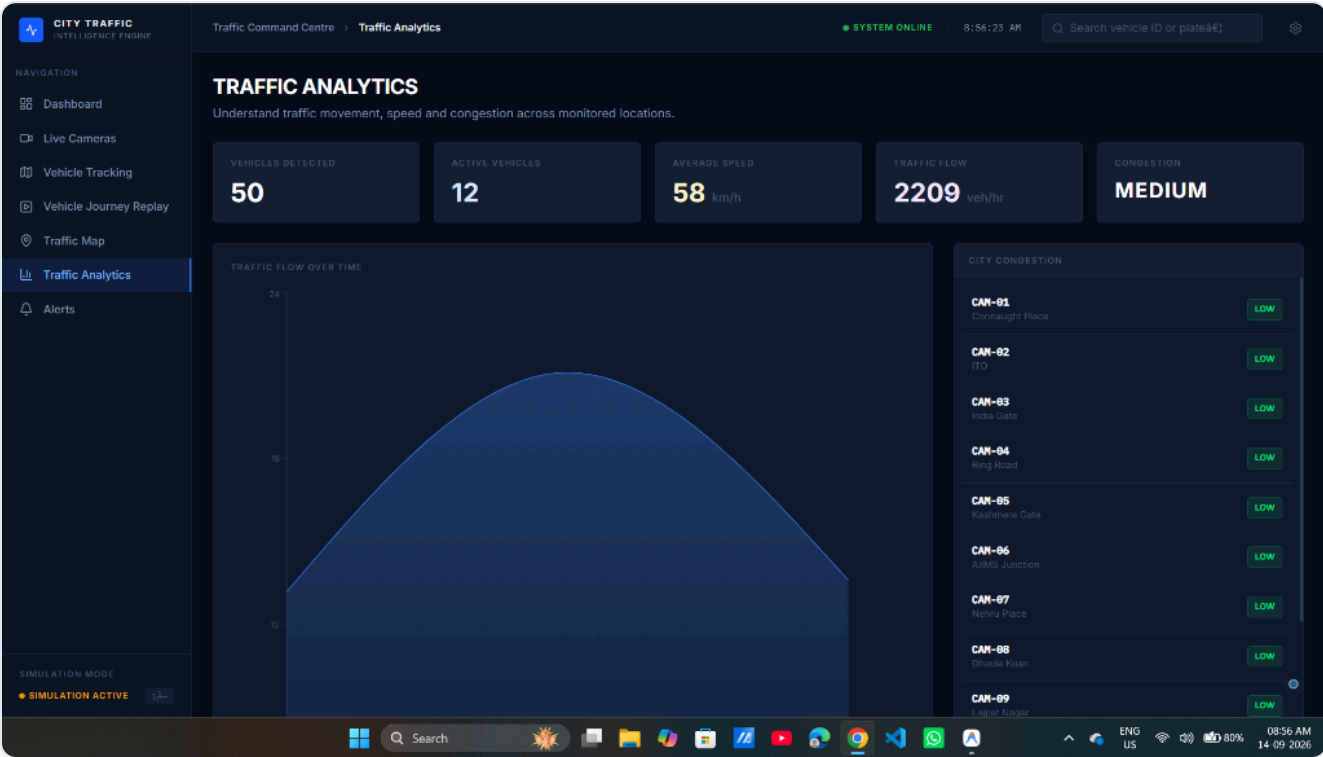
How it works: A continuous background queue evaluates vehicle speeds and route logic. If a vehicle exceeds speed limits or if a specific road's average speed drops drastically, an alert is generated.

The Interface: Capable of buffering up to 500 active alerts. Operators can click on alerts to instantly locate the offending vehicle on the map or mark the situation as resolved.



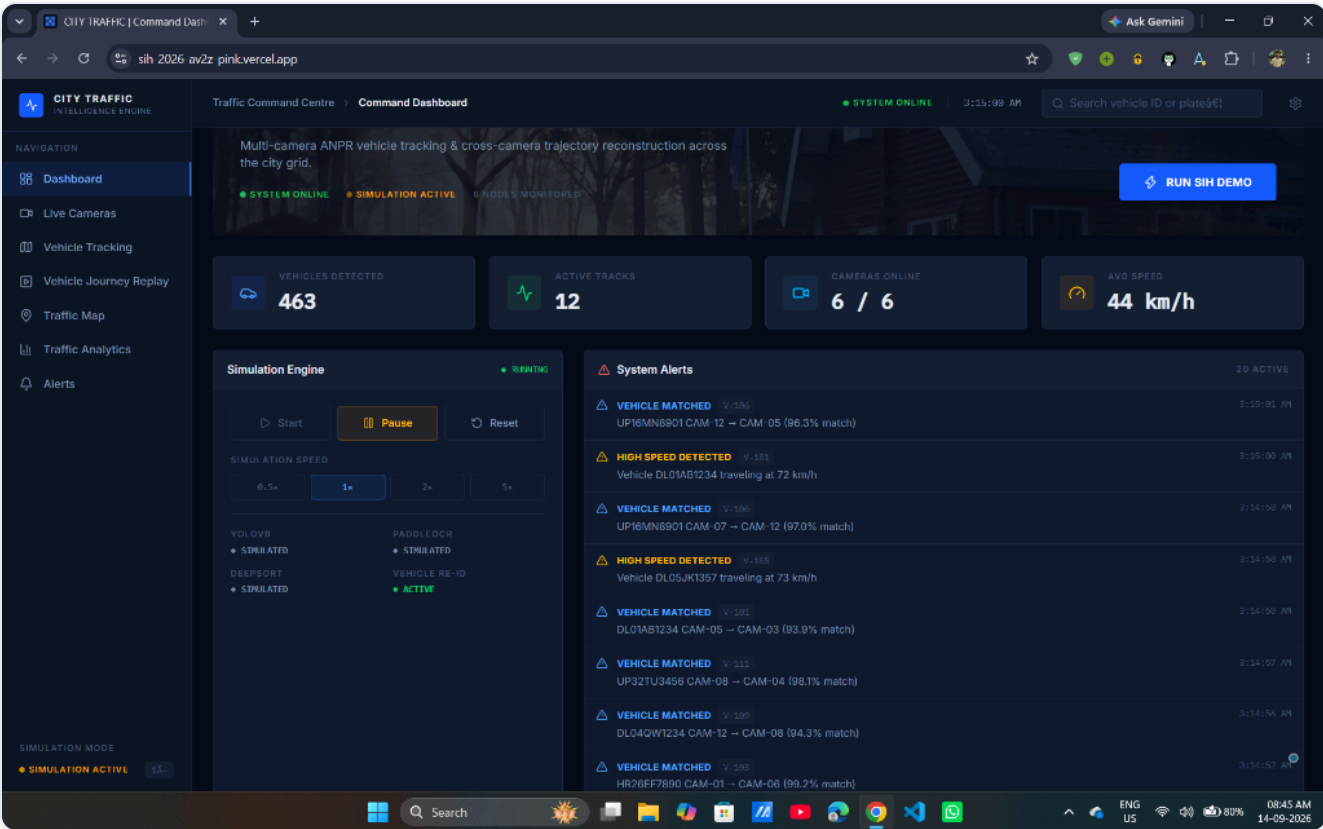
C. Live Traffic Analytics & Congestion Heat

How it works: The system aggregates vehicle counts over time windows to calculate Flow Rate (Vehicles per Hour). By tracking the average speed of all vehicles at a junction, it dynamically classifies the congestion as LOW, MEDIUM, or HIGH.



D. Command Dashboard

Provides a bird's-eye view of active cameras, system-wide total detected vehicles, and unresolved high-priority alerts.



3. Detailed Technology Stack

Our solution utilizes a modern Edge-to-Cloud architecture. Video feeds are analyzed using advanced Computer Vision models, and the metadata is sent to our cloud dashboard.

1. Computer Vision Pipeline (The Brains)

- **YOLOv8 (Object Detection):** Chosen for its industry-leading speed and accuracy. YOLOv8 scans raw camera frames to detect and draw bounding boxes around cars, motorcycles, and trucks in milliseconds.
- **PaddleOCR (ANPR):** Optical Character Recognition optimized for diverse and challenging text. It extracts the license plate numbers from the bounding boxes provided by YOLO.
- **DeepSORT (Vehicle Re-Identification):** Uses Kalman filtering and deep metric learning to track a vehicle across consecutive frames. It ensures that a car waiting at a red light isn't counted multiple times.

2. Frontend (The Command Center)

- **React.js 19 & TypeScript:** Provides a highly interactive, type-safe, and scalable user interface.
- **Tailwind CSS v4:** Used to design the dark, cinematic, enterprise-grade UI rapidly.
- **React-Leaflet & Google Maps:** Renders the highly detailed, interactive city maps and glowing vehicle trajectories without relying on expensive enterprise mapping APIs.

3. Backend & API

- **Python 3.10 & FastAPI:** Handles incoming metadata from the computer vision models. FastAPI is chosen for its extreme asynchronous performance (ASGI).
- **Asyncio Simulation Engine:** For the SIH demonstration, we built a complex tick-based simulation engine in Python that generates realistic traffic data, mimicking a real city's camera network.

4. Pitch & Presentation Guide (For Judges)

- **The Hook:** *"Cities have thousands of cameras, but nobody is watching them. Our system transforms those dumb CCTVs into a smart, proactive city brain."*
- **The Tech:** *"By combining YOLOv8 for detection and DeepSORT for tracking, we create a continuous mesh network. If a suspect vehicle passes one camera, we instantly know its trajectory across the entire city."*
- **The Impact:** *"Traffic police no longer react to congestion—they are proactively alerted to it. Our dashboard provides low-latency, actionable intelligence that saves time, fuel, and lives."*